1 Quiz 1

1.1 Question 1

Determine whether each of the following sets is convex or not, and give brief justifications for your answers.

1.1.1 Part A

$$\{x \mid a^T x \geq b\}$$

where $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$ (we do not forbid a is an all-zero vector).

1.1.2 Part B

$$\{x \mid x^T P x \leq 1\}$$

where $P \in \mathbf{S}^n$.

1.1.3 Part C

$$\{x/y \mid x \in \mathbb{R}^n, y > 0\}$$

1.2 Question 2

State concisely the separating hyperplane theorem in your own words (or in formal math definition). No more than 35 words.

2 Quiz 2

2.1 Question 1

Determine whether each of the following functions is convex or not, and give brief justifications for your answers.

2.1.1 Part A

$$f(x) = a^T x + b, \quad a \in \mathbb{R}^n, b \in \mathbb{R}$$

2.1.2 Part B

$$f(x) = \log \sum_{i=1}^{n} e^{a_i x_i}, \quad a_i \in \mathbb{R}$$

2.1.3 Part C

$$f(x) = x^T A^T Q A x, \quad Q \in \mathbf{S}^m, A \in \mathbb{R}^{m \times n}$$

2.2 Question 2

State concisely the second-order condition for convex functions in your own words (or in formal math definition). No more than 25 words.

2.3 Question 3

In class, we learned that ”minimization can preserve the convexity of a function” under certain conditions. State concisely these conditions (or in formal math definition). No more than 40 words.